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Financial Illiteracy and Pension Contributions: A Field Experiment on Compound Interest in China

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1 Big picture

- **Question.** Does misunderstanding compound interest reduce pension saving, and can targeted financial education raise pension contributions?
- **Setting.** The paper studies the New Rural Social Pension Insurance Program (NRSPIP) in rural China using a field experiment run in one county of Shaanxi province. More than 1,100 households are randomly assigned to a control group, a calculation group, or an education group.
- **Core challenge.** Low contributions to a highly subsidized pension plan could reflect many things: liquidity constraints, procrastination, low trust, weak understanding of the contract, or a specific bias in evaluating compounding. The design needs to separate “people learn that pensions are good” from “people learn how compound interest works.”
- **Main conceptual contribution.** The paper focuses on a narrow and economically important dimension of financial literacy: understanding compound interest. It then links this specific bias to actual administrative pension contributions rather than to survey intentions.
- **Main empirical design.** Households are randomly assigned, before they choose their annual contribution, to one of three interventions:
 - *Control:* explanation of the pension contract plus a survey.
 - *Calculation:* the control treatment plus explicit pension-benefit calculations.
 - *Education:* the calculation treatment plus three compound-interest questions and a direct explanation of the concept of compound interest.
- **Main baseline result.** The education treatment raises annual pension contributions by about 49–53 RMB, which is roughly a 37%–40% increase relative to the control-group mean of 133 RMB. The calculation treatment also raises contributions, but only by about 20–25 RMB.
- **Take-up versus amount.** The intervention has essentially no effect on take-up because participation is already extremely high after the intervention (about 98% in all groups). The effect operates mainly on the intensive margin: how much households contribute.
- **Key mechanism evidence.** The education effect is larger for households that initially underestimate compound interest, and it becomes much larger than the calculation effect for older households. Education also improves post-intervention understanding of compound interest.
- **Structural result.** In the model, households in the control group strongly underestimate compounding, while the education group does so much less. The structural estimates therefore line up with the reduced-form treatment effects.
- **Counterfactual and welfare result.** A financial-education counterfactual raises simulated contributions more than doubling the subsidy does. The education treatment raises lifetime utility by about 8.6%, while eliminating misunderstanding of compound interest entirely raises lifetime utility by about 10.0%.
- **Economic takeaway.** Targeted education that removes a specific cognitive bias can materially increase retirement saving, and it appears more powerful than simply giving households benefit projections or larger subsidies.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper studies the **New Rural Social Pension Insurance Program (NRSPIP)**, introduced in 2009.
- It is a **voluntary defined-contribution pension plan** with unusually large government subsidies.
- In the study county, households can choose one of five annual contribution levels: 100, 200, 300, 400, or 500 RMB.
- The corresponding annual subsidy schedule is 30, 30, 40, 45, and 50 RMB.
- The account earns the official one-year base rate of the People’s Bank of China, which is 2.5% in the study period, with annual compounding.
- After age 60, pension income equals a basic government pension plus the balance in the individual account divided by 139 months.
- The basic pension is 80 RMB per month, or 960 RMB per year.

The monthly benefit formula is

$$\text{monthly benefit} = \text{basic pension} + \frac{\text{individual account balance}}{139}.$$

Interpretation. This is not a low-return saving product. It is a highly subsidized one, so persistent low contributions are more plausibly related to misunderstanding or behavioral frictions than to poor financial incentives.

2.2 How generous the pension is

For a household that starts contributing at age 30 and pays 100 RMB per year for 15 years, the paper’s example implies an annual pension of 1,259 RMB after age 60. Of this amount, about 82% comes from the subsidy and the interest earned on it. At the 500 RMB contribution level, the annual pension in the same example is 2,224 RMB, and even then about 56% comes from subsidy-related funds.

The plan is also generous relative to ordinary bank saving. The paper notes that for a 45-year-old, contributing the maximum amount generates a return roughly 500 basis points higher than a certificate of deposit.

Interpretation. Because the contract is attractive on its face, the paper can plausibly interpret very low realized contributions as a decision problem rather than as a bad-product problem.

2.3 Data source and sample construction

- The county launched the pension plan in 2010. Those 2010 contributions provide the paper’s **baseline administrative data**.

- The field experiment is run during the 2011 registration period.
- Fourteen villages are randomly selected as experimental sites.
- A research team visits households in place of the usual village coordinators and then hands the recorded choices back to the coordinators for actual collection.
- Out of 1,330 visited households, 177 do not answer the door, 32 decline to participate, and 17 are ineligible but mistakenly surveyed.
- The final survey sample contains **1,104 households**.
- Administrative data are collected on actual pension contributions in both 2010 and 2011.
- The paper later also collects **2018 administrative contribution data** to study persistence.

Interpretation. The key dependent variable is not self-reported intended saving. It is the actual contribution later recorded by the pension system.

2.4 Experimental design and treatment arms

Randomization is at the household level. Assignment is determined by interview order along village roads. Enumerators walk door to door and assign households cyclically: first to control, second to calculation, third to education, then back to control. Because the same rule is used for all enumerators, the design naturally lends itself to village fixed effects and enumerator fixed effects.

The three arms are:

- **Control.** The enumerator explains the pension contract and administers a survey on demographics, wealth, family structure, preferences, and financial literacy.
- **Calculation treatment.** The control treatment plus explicit calculations of pension benefits for each contribution level, using the example of contributing from age 30 onward.
- **Education treatment.** The control treatment plus three compound-interest questions, explanation of the correct answers, a short lesson on compound interest, and the same pension-benefit calculations used in the calculation treatment.

Interpretation. The comparison between control and calculation isolates the value of *benefit information*. The comparison between education and calculation isolates the value of *understanding compounding itself*.

2.5 Main variables

Outcome variables.

- **Take-up after intervention:** whether the household contributes to the pension plan in 2011.
- **Contribution level:** actual 2011 contribution in RMB.
- **Change in contribution:** the difference between the 2011 contribution and the 2010 baseline contribution.
- **Long-run contribution:** the contribution recorded in 2018.

Financial-literacy variables.

- During the education treatment, households answer three compound-interest questions that vary the horizon and interest rate.
- In the post-intervention survey, all households answer five financial-literacy questions, including compound-interest, inflation, and numeracy questions.
- The main literacy measure is the **absolute distance** between the chosen answer and the correct answer, normalized by the control-group standard deviation and signed so that a higher value means better understanding.

Other controls.

- Gender, age, years of schooling, household size, land for production, and car ownership enter the reduced-form regressions.
- The survey also elicits risk aversion and patience, which are used in the structural exercise.

2.6 Descriptive facts

The sample is well balanced across the three groups on most baseline covariates. The only clear imbalance is car ownership, which is somewhat lower in the education group. Baseline pension participation in 2010 is already high at 93%, and average baseline contribution is 104 RMB.

After the intervention, take-up is roughly 98% in all groups, but contribution amounts diverge strongly:

- control: 133 RMB,
- calculation: 156 RMB,
- education: 182 RMB.

On financial literacy, misunderstanding of compound interest is severe:

- 56% of households say they do not know the answer to the simplest compound-interest question even after repeated prompting;
- among those who do answer, 73% underestimate compound interest,
- 18% answer correctly,
- and 9% overestimate.

Interpretation. The sample is not one in which households fail to save because the pension is unknown. The contract is familiar, but understanding of compounding is extremely poor.

3 Models and empirical specifications (all equations + interpretation)

3.1 Pension payoff mapping

The pension contract mechanically links today's contribution choice to retirement benefits through annual compounding, government matching, and the post-retirement annuity conversion rule. This matters because the whole paper is about whether households correctly translate those future payoffs into present contribution decisions.

Interpretation. The reduced-form experiment is built on a setting where the mapping from current saving to future income is known, stable, and calculable.

3.2 Benchmark life-cycle model

To rationalize contribution choices, the paper embeds pension saving in a standard life-cycle consumption problem with uncertain lifetime and uncertain income. Individuals maximize expected lifetime utility:

$$E \left[u(C_S) + \sum_{t=S+1}^N \beta^{t-S} u(C_t) \prod_{j=S}^{t-1} p_j \right],$$

subject to

$$X_{t+1} = R(X_t - C_t - Q_t) + Y_{t+1} + Z_{t+1}, \quad X_{t+1} \geq 0.$$

Here:

- C_t is total consumption at age t ,
- Q_t is pension contribution at age t ,
- X_t is cash on hand,
- Y_{t+1} is labor income,
- Z_{t+1} is pension income after retirement,
- p_t is survival probability,
- and β is the discount factor.

Utility has constant relative risk aversion:

$$u(C) = \frac{C^{1-\rho}}{1-\rho}.$$

Interpretation. Pension contributions are one component of intertemporal allocation. The paper does not model the decision in isolation; it nests it inside a standard consumption-saving problem.

3.3 Income uncertainty and Euler equation

Labor income is decomposed into permanent and transitory components:

$$Y_t = P_t U_t, \quad P_t = G_t P_{t-1} N_t,$$

where U_t and N_t are lognormal shocks and G_t is deterministic age-specific income growth. Normalizing by permanent income gives the Euler equation

$$u'(c_t(x_t)) = \beta R p_t E[\nu'(c_{t+1}(x_{t+1})G_{t+1}N_{t+1})].$$

The paper solves this dynamic program for each of the five contribution levels, then compares implied lifetime utility across the contribution options.

Interpretation. The benchmark model generates a normative contribution choice under correct understanding of compounding. It becomes the benchmark against which actual choices are compared.

3.4 Why the benchmark alone is not enough

The calibrated benchmark predicts that households should contribute much more than they actually do. The model-implied mean contribution is about 225 RMB, with a 95% bootstrap confidence interval of 215–236 RMB, while the actual average contribution is only 104 RMB.

Interpretation. Something is missing from the rational benchmark. The paper’s candidate friction is systematic underestimation of compound interest.

3.5 Modeling misunderstanding of compound interest

The paper introduces a parameter θ that captures underestimation of compounding:

$$f(i, t, \theta) = (1 + r)^{(1-\theta)t}.$$

Perceived future value is then

$$FV = PV \cdot f(i, t, \theta).$$

Interpretation of θ :

- $\theta = 0$: unbiased consumer who compounds correctly;
- $0 < \theta < 1$: consumer who underestimates compound interest;
- larger θ : larger underestimation bias.

Interpretation. This is a parsimonious way to distort only the intertemporal-growth mapping while keeping the rest of the life-cycle structure intact.

3.6 Baseline reduced-form treatment regression

The main reduced-form specification is

$$q_{ij} = \alpha_j + \alpha_k + \beta_e T_{ij}^e + \beta_c T_{ij}^c + \phi X_{ij} + \varepsilon_{ij},$$

where:

- q_{ij} is either 2011 pension participation, the 2011 contribution level, or the change in contribution from 2010 to 2011 for household i in natural village j ;

- T_{ij}^e is the education-treatment indicator;
- T_{ij}^c is the calculation-treatment indicator;
- X_{ij} contains socioeconomic controls;
- α_j and α_k are village fixed effects and enumerator fixed effects.

Standard errors are clustered at the level of 93 natural villages.

Interpretation. Because treatment is randomized, β_e and β_c are intent-to-treat estimates. The fixed effects mainly improve precision and absorb village- and enumerator-specific implementation differences.

3.7 Age-heterogeneity specification

To distinguish “learning pension benefits” from “learning compound interest,” the paper interacts treatment with age groups:

$$q_{ij} = \alpha_j + \alpha_k + \sum_{k=2}^4 \beta_k T_{ij}^k + \sum_{k=1}^4 \beta_k^e T_{ij}^{e,k} + \sum_{k=1}^4 \beta_k^c T_{ij}^{c,k} + \phi X_{ij} + \varepsilon_{ij}.$$

The four age groups are 25–35, 35–45, 45–55, and 55–60.

The logic is simple:

- The calculation treatment shows benefits for someone who starts saving at age 30.
- Someone already near age 30 can directly map that information into her own choice.
- Older individuals must do an extra conceptual step: translate the age-30 example into their own shorter contribution horizon.
- The education treatment should therefore dominate the calculation treatment more strongly for older people if understanding compounding is the true mechanism.

Interpretation. This is one of the paper’s sharpest mechanism tests.

3.8 Financial-literacy regressions

The same treatment logic is used for post-intervention financial literacy. The dependent variable becomes the absolute distance to the correct answer for each question. The paper estimates a seemingly unrelated regression (SUR) across the five literacy questions and also forms average standardized treatment effects following Kling, Liebman, and Katz (2007).

Interpretation. If the education treatment mainly works by making the pension look attractive in a vague way, it should improve broad financial-confidence measures. If it works by teaching compounding, the improvement should be concentrated on compound-interest questions.

3.9 Structural estimation

The structural exercise uses both survey answers and actual contribution choices. The paper estimates two versions:

- **Specification 1:** a four-parameter GMM-style estimation of ρ , β , θ_1 (control-group bias), and θ_2 (education-group bias).
- **Specification 2:** a random-coefficients MSM-style estimation in which the compounding-bias parameters are lognormally distributed with means μ_1 and μ_2 for the control and education groups and common standard deviation σ .

Both specifications are estimated using 25 moment conditions: risk-attitude moments, time-preference moments, compound-interest moments, and moments from actual pension contributions in the control and education groups.

Interpretation. The structural model is used to quantify how much the intervention changes the bias parameter and to conduct policy counterfactuals, not to identify the reduced-form treatment effect itself.

3.10 Counterfactual policy simulations

The paper simulates three scenarios:

- **Baseline:** households behave like the control group.
- **Subsidy policy:** the existing subsidy is doubled.
- **Financial education policy:** the compounding bias is reduced to the level estimated for the education group.

It then computes both predicted pension contributions and lifetime utility under each scenario.

Interpretation. The model-based counterfactuals translate the reduced-form experiment into a policy comparison: is it better to pay households more or to make them understand the contract better?

4 Identification and instruments (what makes the estimates credible)

4.1 Main source of causal identification

There is no traditional instrument in the reduced-form analysis. The paper’s causal leverage comes from **randomized household assignment** before the annual contribution decision and from the use of **administrative outcome data** after the intervention.

Interpretation. Credibility comes from experimental design, not from external instruments.

4.2 Why the comparison across groups is informative

The paper is not just a treatment-versus-control design. It is a three-arm mechanism design:

- control vs. calculation measures the effect of explicit pension-benefit information;
- calculation vs. education measures the extra effect of explaining compound interest itself;

- control vs. education captures the full effect of the targeted financial education package.

Interpretation. This is what lets the paper move beyond “financial education works” toward a more specific statement: *teaching compounding* matters more than simply showing benefit projections.

4.3 Balance and nonresponse

The overall nonresponse rate is 17%, but differences in nonresponse across groups are not statistically significant. Baseline covariates are mostly balanced across the three groups. Car ownership differs somewhat, but the main results are robust to the inclusion of standard controls.

Interpretation. Randomization appears to work well in practice.

4.4 Implementation and actual payment

The experiment is run inside the normal administrative process. Households first register a contribution choice and later make the payment to village coordinators. Only about 1% of households fail to adhere to the level they originally chose.

Interpretation. The study captures real decisions in a naturally occurring setting rather than hypothetical survey responses.

4.5 Why reverse causality is weak

The treatment is delivered before households make the 2011 contribution choice, and the paper uses actual contributions recorded after the intervention. Moreover, when the dependent variable is the change from 2010 to 2011, the coefficients remain almost unchanged.

Interpretation. The main results are not driven by preexisting differences in contribution trends across groups.

4.6 Village and enumerator effects

Assignment cycles along village roads under the supervision of village coordinators. Because the sequence rule is identical for all enumerators, the paper includes:

- natural-village fixed effects,
- and enumerator fixed effects.

Interpretation. These controls absorb location-specific and interviewer-specific implementation differences that could otherwise contaminate the treatment comparisons.

4.7 Enumerator persuasion and differential effort

A potential concern is that enumerators may simply have pushed education-group households harder to contribute, or may have spent more effort on participants who looked especially biased.

The paper argues against this in several ways:

- enumerators had no incentive to induce higher contributions;
- they were explicitly instructed not to pressure households;
- stronger persuasion cannot explain why the education group later performs better on compound-interest questions;
- survey duration is roughly 28–30 minutes regardless of the participant’s initial bias, which suggests no systematic extra effort for more-biased respondents.

Interpretation. The evidence points toward real learning rather than interviewer intensity.

4.8 Why the mechanism evidence is convincing

The paper’s preferred interpretation is that education increases contributions by improving understanding of compound interest. Three pieces of evidence line up in that direction:

- the treatment effect is larger for those who initially underestimate compounding;
- the education treatment dominates the calculation treatment especially for older households, who need to translate a generic age-30 example into their own case;
- and the education treatment improves answers to compound-interest questions more than to unrelated numeracy or inflation questions.

Interpretation. The paper cannot perfectly rule out every alternative story about salience or credibility, but the full pattern fits the compounding mechanism much better than a general-information story.

4.9 Persistence and robustness

The paper also shows that the treatment effects persist in the long run. Using 2018 administrative data, the education treatment still raises contributions by about 45 RMB relative to the control group, and the calculation treatment by about 21 RMB. Only about 9% of participants change their contribution in the long run.

Interpretation. The treatment is not a one-shot survey effect. It changes saving behavior persistently, consistent with inertia or habit once households choose a higher contribution schedule.

4.10 What the paper can and cannot distinguish

What the paper *can* establish:

- misunderstanding compound interest is causally related to low pension contributions in this setting;
- targeted education about compounding raises contributions;
- part of the treatment effect operates through improved understanding rather than generic pension information.

What the paper *cannot* fully establish:

- whether pension saving crowds out or crowds in other forms of saving;
- whether the same magnitudes apply in settings with weaker baseline awareness or with longer delays between commitment and payment;
- whether the structural welfare gains are robust to materially different life-cycle assumptions.

5 Tables and figures

5.1 Table 1 and Figure 1: pension contract and experimental design

Read:

- Table 1 shows that households choose among five annual contribution levels, from 100 RMB to 500 RMB, with annual subsidies between 30 RMB and 50 RMB.
- The table's worked example shows that a 30-year-old contributing 100 RMB per year for 15 years receives 1,259 RMB annually after age 60, while a 500 RMB contributor receives 2,224 RMB.
- Figure 1 lays out the three-arm experiment cleanly: control ($N = 372$), calculation ($N = 363$), and education ($N = 369$), all followed by the same post-treatment survey and then actual take-up and contribution decisions.

Economic message. The setting combines a generous pension contract with a clean experimental design that separates “information about the pension” from “understanding compound interest.”

from the non-rural, urban example, for those who are already 60 or older, all their eligible children living in the same village participate in the program, the parents can receive the basic pension every month without making any

Table 1					
Pension contract					
A. Pension subsidy					
Options	Contribution level (RMB/year)	Percentage of annual per capita income		Government subsidy (RMB/year)	
1	100	1.5%		30	
2	200	3.1%		30	
3	300	4.6%		40	
4	400	6.2%		45	
5	500	7.7%		50	
B. Example of pension benefit					
Age when you start to contribute	30				
Annual contribution level (RMB/year)	100	200	300	400	500
Annual subsidy (RMB/year)	30	30	40	45	50
A: Basic pension after age 60 (RMB/year)	960	960	960	960	960
B: Amount from individual account balance (RMB/year)	299	529	781	1,023	1,264
C=A+B: Amount received annually after age 60 (RMB/year)	1,259	1,489	1,741	1,983	2,224

Panel A shows the corresponding government subsidy for each contribution level in the pension plans. Panel B provides an example to illustrate the explicit benefit of each contribution level for one who starts to contribute at age 30 and contributes for the next 15 years. The interest rate is assumed to be 2.5%, which was the 1-year interest rate in China at the time of this study.

(a) Table 1

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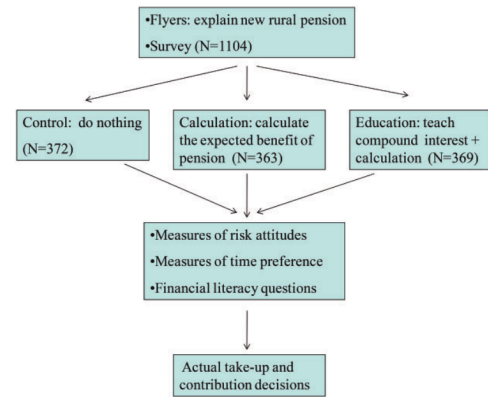


Figure 1
Experimental design

(b) Figure 1

Figure 1: Pension contract and experimental design.

5.2 Table 2 and Figure 2: sample balance and initial misunderstanding

Read:

- Table 2 shows that the treatment groups are broadly balanced at baseline. Baseline take-up is about 93%, and baseline average contribution is about 104 RMB.
- The post-intervention mean contribution is 133 RMB in the control group, 156 RMB in the calculation group, and 182 RMB in the education group.
- Figure 2 shows severe ex ante misunderstanding of compound interest. Among respondents who answer the question, the large majority underestimate future value; only 18% answer correctly.

Economic message. The paper starts from a setting with poor understanding of compounding but with no major baseline imbalance across treatment arms.

Table 2
Summary statistics

	Total	Control	Calculation treatment	Education treatment	p-value
<i>A. Baseline</i>					
Male	0.67 (0.47)	0.64 (0.48)	0.70 (0.46)	0.67 (0.47)	.22
Age	44.90 (9.18)	44.87 (9.66)	44.40 (9.00)	45.42 (8.84)	.30
Years of schooling	8.69 (2.50)	8.71 (2.56)	8.67 (2.56)	8.70 (2.40)	.97
Household size	4.78 (1.34)	4.80 (1.37)	4.82 (1.38)	4.73 (1.29)	.66
Land for production	3.75 (1.61)	3.75 (1.66)	3.76 (1.59)	3.73 (1.57)	.98
Own business	0.14 (0.34)	0.16 (0.36)	0.12 (0.32)	0.13 (0.34)	.32
Own a car	0.10 (0.30)	0.13 (0.34)	0.11 (0.31)	0.06 (0.24)	.00
Own a motorcycle	0.44 (0.50)	0.48 (0.50)	0.44 (0.50)	0.40 (0.49)	.09
Saving for children	0.81 (0.39)	0.79 (0.40)	0.83 (0.38)	0.80 (0.40)	.50
Saving for future when she/he is old	0.25 (0.44)	0.26 (0.44)	0.26 (0.44)	0.25 (0.43)	.92
Have a private pension plan	0.13 (0.34)	0.14 (0.35)	0.13 (0.33)	0.13 (0.33)	.83
Receiving reimbursement from government health insurance	0.91 (0.28)	0.93 (0.25)	0.90 (0.31)	0.91 (0.28)	.53
Baseline take-up	0.93 (0.26)	0.92 (0.28)	0.93 (0.26)	0.94 (0.23)	.34
Baseline contribution level	104.17 (65.28)	104.57 (71.23)	106.34 (70.03)	101.63 (53.14)	.57
<i>B. Post-intervention</i>					
Risk aversion	4.04 (1.68)	4.11 (1.65)	3.98 (1.71)	4.03 (1.69)	.56
Patience	2.82 (2.61)	2.64 (2.64)	2.86 (2.61)	2.95 (2.59)	.26
Absolute distance to the correct answer to question 1	-1.08 (0.99)	-1.13 (1.00)	-1.06 (0.98)	-1.07 (0.99)	.58
Absolute distance to the correct answer to question 2	-1.72 (1.00)	-1.73 (1.00)	-1.74 (0.97)	-1.68 (1.01)	.72
Absolute distance to the correct answer to question 3	-2.95 (1.09)	-3.05 (1.00)	-2.92 (1.20)	-2.89 (1.13)	.06
Absolute distance to the correct answer to question 4	-0.80 (0.99)	-0.82 (1.00)	-0.790 (0.99)	-0.78 (0.99)	.78
Absolute distance to the correct answer to question 5	-1.55 (1.00)	-1.59 (1.00)	-1.53 (0.98)	-1.53 (1.01)	.59
Take-up	0.98 (0.13)	0.98 (0.13)	0.98 (0.14)	0.99 (0.12)	.83
Contribution level	157.16 (123.72)	133.06 (96.62)	156.19 (125.19)	182.38 (140.80)	.00
Observations	1,104	372	363	369	

Standard deviations are in parentheses. Baseline take-up and baseline contribution are the actual pension participation and contributions before my intervention in 2010. Absolute distance to the correct answer to questions 1–5 is normalized by standard deviation of the control group. *p*-values are for Wald test of equal means of three groups. **p* < .1; ***p* < .05; ****p* < .01.

(a) Table 2

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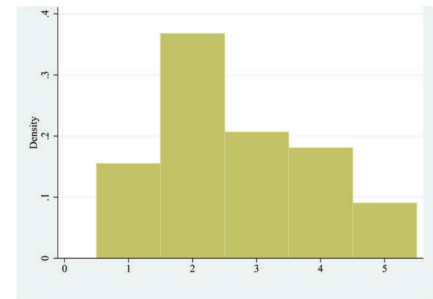


Figure 2
Response to compound interest question
This figure shows the distribution of responses to the compound interest rate question before intervention. I ask subjects the following question: You deposit 100 RMB as a certificate of deposit this year at a constant interest rate of 9% per year. Interest is compounded annually. How much money could you receive in 30 years? (1) Less than 300; (2) 300–500; (3) 500–1,000; (4) 1,000–1,500; or (5) more than 1,500? 1, 2, 3, underestimate; 4, correct; 5, overestimate.

(b) Figure 2

Figure 2: Summary statistics and pre-intervention misunderstanding of compound interest.

5.3 Figure 3, Figure 4, and Table 4: the main treatment effects

Read:

- Figure 3 shows virtually no treatment effect on take-up: all three groups participate at about 98%.
- The same figure shows a clear contribution effect: the control mean is 133 RMB, the calculation mean 156 RMB, and the education mean 182 RMB.
- Figure 4 shows that after the intervention, the education group is noticeably more likely to choose 300 RMB and 500 RMB contribution levels.
- Table 4 quantifies these differences. The education treatment raises the 2011 contribution by

53.06 RMB, while the calculation treatment raises it by 25.22 RMB.

- Using the change from 2010 to 2011, the corresponding coefficients are 54.57 RMB and 22.34 RMB.
- There is no meaningful treatment effect on take-up, and interactions with baseline contribution are not significant.
- In the heterogeneity column, the education effect is about 56 RMB for those who cannot answer the compound-interest question, but it is 78 RMB smaller for overestimators.

Economic message. The intervention works mainly on the intensive margin and especially for households who initially misunderstand compounding.

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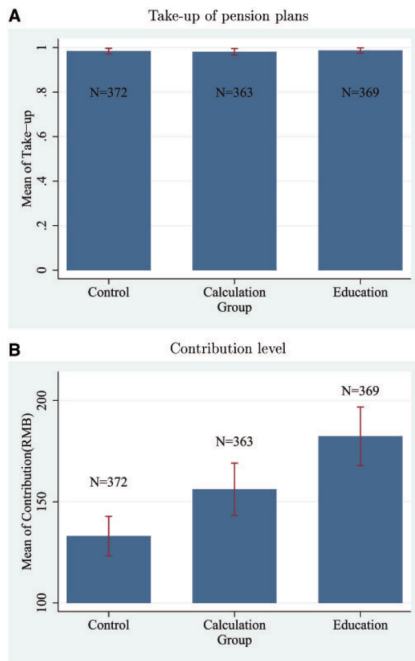


Figure 3
Treatment effects
Panel A shows the treatment effect on take-up of pension plans. In the control group, take-up is 98.4%. In the calculation group, take-up is 98.1%. In the education group, take-up is 98.6%. This suggests that almost all the households in the three groups participate in the pension plans. Panel B shows the treatment effect

(a) Figure 3

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Table 4
Effect of education and calculation on contribution level

Dep. var:	Individual adoption of pension	Individual contribution level of pension	OLS regression Change in contribution level of pension All sample	Individual contribution level of pension	Control and education
Sample:	1	2	3	4	5
Education	0.004 (0.009)	53.06 (9.28)***	54.57 (8.94)***	68.71 (14.85)***	55.6 (10.87)***
Calculation	-0.002 (0.009)	25.22 (9.24)***	22.34 (8.13)***	18.68 (12.97)	
Education*Overestimation (option 5)					-78.16 (28.81)***
Education*Correct estimation (option 4)					-26.19 (47.85)
Education*Underestimation (option 3)					-3.03 (53.39)
Education*Underestimation (option 2)					-33.01 (36.43)
Education*Underestimation (option 1)					14.11 (37.17)
Education*Baseline contribution				-0.14 (0.11)	
Calculation*Baseline contribution				0.04 (0.10)	
Baseline contribution				0.74 (0.08)***	
Wald test: $\beta_1 = \beta_2$					
p-value	.5009	.0064***	.0004***		
Obs.	1,104	1,104	1,104	1,104	741
Omitted treatment			Control		
Mean of dep. var. for omitted treatment:	0.0984	133.06	28.49	133.06	133.06
Fixed effects for village and enumerator	Y	Y	Y	Y	Y
R-squared	0.0648	0.0895	0.0963	0.2295	0.1546

Standard errors are clustered by ninety-three natural villages. Robust standard errors are clustered in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$. In Column 1, the dependent variable is individual pension participation. In Columns 2, 4, and 5, the dependent variable is individual contribution level after my intervention in 2011. In Column 3, the dependent variable is changes in individual contribution level from 2010 to 2011. In Column 5, I restrict the sample to the control group and the education group only. β_1 is the coefficient of the education treatment and β_2 is the coefficient of the calculation treatment.

those in Column 2. In Column 4, I study the heterogeneous effect with respect to the baseline contribution in 2010. Coefficients of the interaction between

(b) Table 4

Figure 3: Main treatment effects on pension contributions.

5.4 Table 5 and Figure 5: age heterogeneity

Read:

- Table 5 shows that for households aged 25–35, the education and calculation treatments have similar effects.
- For households aged 45–55, the education effect is about 50–62 RMB while the calculation effect is only about 8–16 RMB; the difference is statistically significant.

- For the coarser 40–60 age group, the gap between education and calculation is about 37–41 RMB and highly significant.
- Figure 5 makes the pattern transparent: the education line is relatively flat across age, whereas the calculation line falls sharply with age.

Economic message. This is the paper’s cleanest mechanism test. Older households need conceptual understanding to translate an age-30 projection into their own shorter saving horizon, so education matters more for them than pure calculation.

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Table 5
Effect of education and calculation on contribution level by age group
Specification:
Dep. var.:

Sample:	OLS regression	
	Individual contribution level	Change in individual contribution level
	All sample	
	1	2
A. Four age groups		
Effect of the education treatment for those between age 25 and 35 (β_{11})	51.89 (26.47)*	37.48 (20.90)*
Effect of the calculation treatment for those between age 25 and 35 (β_{11})	40.76 (25.79)	35.68 (22.60)
Effect of the education treatment for those between age 35 and 45 (β_{12})	60.88 (16.61)***	65.52 (16.83)**
Effect of the calculation treatment for those between age 35 and 45 (β_{12})	46.07 (14.41)***	41.35 (13.06)***
Effect of the education treatment for those between age 45 and 55 (β_{13})	50.20 (15.65)***	61.70 (14.19)***
Effect of the calculation treatment for those between age 45 and 55 (β_{13})	17.79 (12.70)	16.49 (8.81)*
Effect of the education treatment for those between age 55 and 60 (β_{14})	49.62 (26.41)*	42.09 (23.40)*
Effect of the calculation treatment for those between age 55 and 60 (β_{14})	5.71 (21.93)	-13.65 (15.81)
Prob > F(Education=Calculation) for those between age 25 and 35	0.6982	0.9460
Prob > F(Education=Calculation) for those between age 35 and 45	0.4358	0.1947
Prob > F(Education=Calculation) for those between age 45 and 55	0.0136**	0.0030***
Prob > F(Education=Calculation) for those between age 55 and 60	0.1179	0.0269**
B. Two age groups		
Effect of the education treatment for those between age 20 and 40	55.34 (18.03)***	49.11 (17.01)***
Effect of the calculation treatment for those between age 20 and 40	43.16 (17.34)**	35.35 (16.97)**
Effect of the education treatment for those between age 40 and 60	50.05 (11.46)***	55.25 (11.18)***
Effect of the calculation treatment for those between age 40 and 60	13.93 (10.49)	14.54 (8.10)*
Prob > F(Education=Calculation) for those between age 20 and 40	0.5065	0.4406
Prob > F(Education=Calculation) for those between age 40 and 60	0.0077***	0.0007***

Standard errors are clustered by ninety-three natural villages. Robust standard errors are clustered and in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$. Panel A reports the results for four age groups. Panel B reports the results for two age groups as a test for robustness. In Column 1, the dependent variable is the individual level contribution. In Column 2, the dependent variable is the change in the individual level contribution.

(a) Table 5

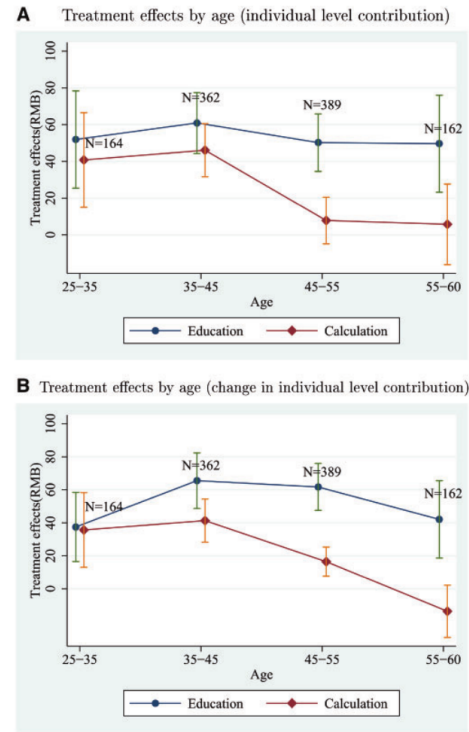


Figure 5

Treatment effects by age

Panel A shows treatment effects for the education and calculation groups by age group. The dependent variable is individual level contribution. The horizontal axis represents four age groups. The vertical axis is the treatment effects. I report one standard error for each coefficient. Panel B shows the treatment effects for the education and calculation groups by age group. The dependent variable is the change in individual level contribution. The horizontal axis represents four age groups. The vertical axis is the treatment effects. I report one standard error for each coefficient.

(b) Figure 5

Figure 4: Age heterogeneity in education and calculation treatment effects.

5.5 Figure 6 and Table 6: post-intervention financial literacy

Read:

- Figure 6 shows that after the intervention, the education group has fewer extremely wrong answers and more correct answers on the post-intervention compound-interest question.
- Table 6 shows that the only single financial-literacy question with a clearly significant education effect is question 3, the compound-interest question.

- The average standardized treatment effect is positive and significant for the compound-interest-question bundle, but not for the non-compound-interest bundle.
- On all five questions together, the average standardized treatment effect is positive and only marginally significant.

Economic message. The intervention changes the specific concept it was designed to teach. That is exactly the pattern we should observe if understanding compound interest is the main causal channel.

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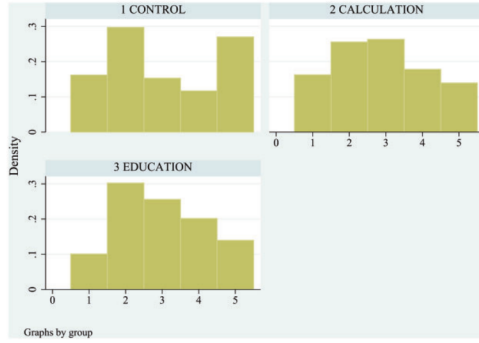


Figure 6
Response to compound interest question
This figure shows the distribution of responses to the compound interest question after intervention. I asked subjects the following question: You deposit 100 RMB as a certificate of deposit this year at a constant interest rate of 6% per year. Interest is compounded annually. How much money could you receive in 30 years? (1) Less than 300, (2) 300–400, (3) 400–500, (4) 500–600, or (5) more than 600? Figures only include those who answered the question and exclude those who did not know. 1, 2, 3, underestimate; 4, correct; 5, overestimate.

(a) Figure 6

Table 6
Effect of the education and calculation interventions on financial literacy

Specification:	SUR regression					Average standardized effect on absolute distance to the correct answer		
	Absolute distance to the correct answer					Question 2 Question 1, 4 and 3 All questions and 5		
	Question 1	Question 2	Question 3	Question 4	Question 5			
Dep. var:								
Sample:	1	2	3	4	5	All sample		
Education	0.078 (0.067)	0.062 (0.009)	0.186 (0.074)**	0.053 (0.068)	0.061 (0.060)	0.124 (0.063)**	0.064 (0.049)	0.088 (0.048)*
Calculation	0.071 (0.067)	-0.011 (0.069)	0.120 (0.074)	0.029 (0.068)	0.043 (0.060)	0.055 (0.057)	0.048 (0.046)	0.051 (0.045)
Obs.	1,104	1,104	1,104	1,104	1,104	1,104	1,104	1,104
Omitted treatment					Control			
Mean of dep. var. for omitted treatment	-1.13	-1.73	-3.06	-0.83	-1.60			
Social-economic variables	Y	Y	Y	Y	Y	Y	Y	Y
Fixed effects for village and enumerator	Y	Y	Y	Y	Y	Y	Y	Y
R-squared	0.1752	0.1550	0.1632	0.1446	0.1545			

In Columns 1 to 5, the dependent variables are the absolute distance between the chosen answers and the correct answers for questions 1 to 5, normalized by the standard deviation of the control group. Column 6 reports average standardized treatment effects on questions 2 and 3, of which both are compound interest questions. Column 7 reports average standardized treatment effects on questions 1, 4 and 5, of which none of them are related to compound interest. Column 8 reports average standardized treatment effects on all questions.

(b) Table 6

Figure 5: Post-intervention understanding of compound interest.

5.6 Table 7: structural estimation and counterfactuals

Read:

- In specification 1, the estimated risk-aversion coefficient is 1.54 and the discount factor is 0.94.
- The control-group compounding-bias parameter is 0.67, while the education-group parameter is 0.09.
- In the random-coefficients specification, the implied mean exponential-growth bias is about 0.50 in the control group and 0.07 in the education group.
- The counterfactual average contribution is 155 RMB in the baseline, 158 RMB when the subsidy is doubled, and 177 RMB under the financial-education policy.
- The paper further reports that lifetime utility rises by 8.6% when households move from the control-group bias level to the education-group bias level, and by 10.0% under perfect understanding.

Economic message. The structural exercise validates the reduced-form story and shows that correcting the bias is quantitatively more effective than making the subsidy more generous.

Table 7
Structural estimation

Parameter	Symbol	(1)	(2)
A. Random coefficients			
Risk aversion	ρ	1.54 (0.032)	1.64 (0.019)
Discount factor	β	0.94 (0.034)	0.95 (0.083)
Exponential growth bias in the control group	θ_1	0.67 (0.001)	
Exponential growth bias in the education group	θ_2	0.09 (0.001)	
Mean of the logarithm of exponential growth bias in the control group	μ_1		-0.04 (0.012)
Mean of the logarithm of exponential growth bias in the education group	μ_2		-2.04 (0.015)
Standard deviation of the logarithm of exponential growth bias	σ		1.07 (0.022)
B. Distribution of coefficients			
Mean of exponential growth bias in the control group			0.50
Standard deviation of exponential growth bias in the control group			0.25
Mean of exponential growth bias in the education group			0.07
Standard deviation of exponential growth bias in the education group			0.03
C. Counterfactual simulation		Pension contribution (RMB)	
Baseline			155
Increase the subsidy by 100%			158
Financial education			177

I use both survey data and actual pension contributions from the experimental data to estimate risk aversion parameters, time preference parameters, and parameters of underestimating compound interest. Column 1 estimates four parameters: the coefficient of relative risk aversion, ρ ; the discount factor, β ; the magnitude of underestimating compound interest in the control group, θ_1 ; and the magnitude of underestimating compound interest in the education group, θ_2 . Column 2 assumes that the parameters of underestimating compound interest are lognormally distributed in both the education group and the control groups and estimate five parameters. μ_1 and σ are the mean and standard deviation of the logarithm of exponential growth bias in the control group, and μ_2 and σ are the mean and standard deviation of the logarithm of exponential growth bias in the education group. Panel A reports the estimated coefficients. Standard errors are reported in parentheses. Panel B reports the distribution of coefficients by taking draws from the coefficients in panel A. Panel C reports the simulated pension contribution based on three counterfactual scenarios. The first scenario is the baseline when all subjects underestimate the benefit of compound interest. In the second scenario, I consider the subsidy policy that increases the existing

Figure 6: Table 7 from the paper: structural estimation and policy counterfactuals.

6 Short summary

- The paper studies whether misunderstanding compound interest reduces pension contributions in rural China.
- It runs a three-arm field experiment that cleanly separates pension-benefit information from education about the concept of compound interest.
- The education treatment raises annual contributions by about 49–53 RMB, or roughly 37%–40% relative to the control mean; the calculation treatment raises them by only about half as much.
- The strongest mechanism evidence is that the education effect is larger for initial underestimators, larger than the calculation effect for older households, and concentrated on compound-interest literacy rather than on unrelated literacy questions.

- The effect persists in long-run administrative data, suggesting it is not a transient survey artifact.
- Structurally, the education group exhibits much smaller exponential-growth bias than the control group.
- In counterfactuals, financial education is more effective than doubling the subsidy and produces sizable welfare gains.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that misunderstanding compound interest is an economically important barrier to retirement saving in this rural Chinese pension setting. Explaining compound interest causally increases pension contributions, and the effect is larger than the effect of merely showing benefit projections.

7.2 Economic intuition

Households do not just need to know that a pension is beneficial. They need to understand how today's contribution compounds into future retirement income. If they systematically linearize or understate that growth, they will undercontribute even to a highly subsidized plan. Education that corrects this specific bias changes the perceived return to saving and therefore raises contributions.

7.3 Takeaway

The paper's main lesson is that *specific* financial education can be much more powerful than generic information or broader subsidies. In this case, teaching one concept—compound interest—materially improves behavior and welfare.